Unit 1: Number Patterns

Introduction to Number Patterns:

Number patterns have been a topic of interest for humans for thousands of years. You have been exposed to number patterns since a young age. In earlier grades you saw patterns in the form of pictures and numbers. Patterns are repetitive sequences and can be found in nature, shapes, events, sets of numbers and almost everywhere you care to look.

A ***Sequence*** is an ordered set of numbers (called **terms**) that are listed and separated by semi-colons.

e.g.1 1;3;5;7;9;11…

e.g.2 1;2;4;7;11…

e.g.3 5;10;15;20;....

e.g.4 1;4;9;16;25;...

Each number in the sequence has both a _____ and a _____within the sequence.

The _____ within a sequence refers to the numeric value of the term and is denoted by T_n

The _____ within the sequence refers to the position in the sequence and is denoted by ***n***.

e.g. in the third sequence above:

$T_1 = 5$	The value of the 1st term is 5.
$T_2 = 10$	The value of the 2nd term is 10
$T_3 = 15$	The value of the 3rd term is 15.
⋮	
$T_n = 5n$	The value of the nth term (any term) is five times its position. This is the _____ for the number pattern.

A sequence that has a limited number of terms is called a _____

A sequence that has unlimited terms (or has no end) is called an _____

A _____ is a statement in words describing the relationship between the numbers and their position in the sequence. A conjecture should always include the starting point of the sequence.

A _____is a mathematical formula that can be used to determine the value of any term in the sequence. By convention, we find a rule where the position determines the value of the term.

Linear number patterns (Arithmetic sequence):

A ***Linear*** pattern is any pattern with a common **first difference** between consecutive terms. To progress to the next term of a linear number pattern, the common difference is added to the current term.

The ***general term or rule*** for any linear pattern is:

Example 1:

Consider the following pattern.

6;10;14;18;...

(a) What is the constant first difference?

(b) Determine the general rule for this pattern.

(c) Determine the 1 000th term.

(d) Which term of the pattern equals 202?

Example 2:

Consider the following pattern.

5;7;9;11;...

(a) What is the constant first difference?

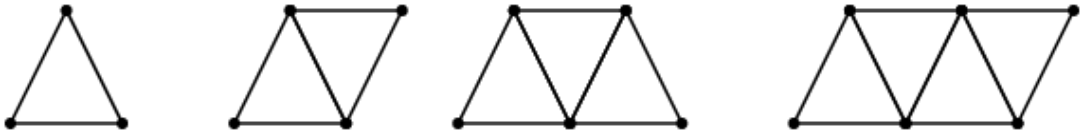
(b) Determine the general rule for this pattern.

(c) Determine the 500th term.

(d) Which term of the pattern equals 103?

Example 3:

Chains of triangles can be built with matches as follows:



(a) How many matches will be used to create a chain of 5 triangles?

(b) Make a conjecture for calculating the number of matches is a chain of *n* triangles.

(c) Determine the general rule for this pattern.

(d) Now determine how many matches will be needed to build a chain of 100 triangles.

Homework: Ex. 1 pg. 88 – 89 (Mind Action Series new edition, 2022).

Arithmetic Sequences with Variable Terms and Miscellaneous Examples:

Example 4:

a) Determine the value of x that makes $18 - x$; $3 + 2x$; $6 + 3x$ an arithmetic sequence.
b) Determine the first term of the sequence if $18 - x$ is the 20th term of the sequence.

Example 5:

Consider the first three terms of an arithmetic pattern:

$$18 - x \text{ ; } 18 - 2x \text{ ; } 18 - 3x \text{ ; } \ldots$$

a) What values can x assume so that the pattern will be increasing?
b) Calculate the common difference if the value of the first term is half that of the third term.

Homework: Ex. 2 pg. 89 – 90 (Mind Action Series new edition, 2022)

Example 6 (Simultaneous equations could be used, see §3):

Calculate the general term of the linear patters with a 6th term of -2 and a 10th term of -18.

Example 7:

The word CLOWN over and over again as follows: CLOWNCLOWNCLOWNCLOWN……

a) What is the 22nd letter?

b) Find the 500th letter.

c) The first W is in the 4th position, the second W is in the 9th position, and so forth. Determine in what position the 100th W will be?

Homework: Ex. 3 pg. 91 – 92 (Mind Action Series new edition, 2022)
Homework: Consolidation and extension exercise pg. 93 – 95